

ABE REVIEW

Q&A IN AGRICULTURAL SCIENCES

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Easy Questions

1. The practice of growing crops or raising animals.

- A. Agriculture B. Crop science C. Animal science D. Crop protection

Answer: A

2. The study of agricultural issues and challenges, including resource use consistent with sustainable production of food, feed, fiber and fuel, including the production, processing and marketing of high quality food and non-food products.

- A. Agriculture B. Crop science C. Horticulture D. Crop management

Answer: B

3. Concerned with the science and business of producing domestic livestock species, including but not limited to beef cattle, dairy cattle, horses, poultry, sheep, and swine.

- A. Animal husbandry B. Livestock science C. Animal science D. Animal protection

Answer: C

4. The general method or the practice of protecting the crop yields from different agents including pests, weeds, plant diseases, and other organisms that cause damage to the agricultural crops.

A. Pest control B. Pest management C. Pest science D. Crop protection

Answer: D

5. The study of soil as a natural resource on the surface of the Earth including soil formation, classification and mapping; physical, chemical, biological, and fertility properties of soils; and these properties in relation to the use and management of soils.

A. Soilogy B. Soil science C. Earth Science D. Soil management

Answer: B

Moderate Questions

6. How many mango seedlings are needed to be purchased for 1 hectare area if the planting density is 10 meters by 10 meters? Give 5% allowance in the number of seedlings purchased.

A. 100 B. 105 C. 150 D. 525

Answer: B

Solution:

$$\begin{aligned}\text{No. of Seedlings} &= \text{Safety Factor} [\text{Area in ha} \times 10,000 \text{ m}^2/\text{ha} / \text{Planting Density in m} \times \text{m}] \\ &= 1.05 [1 \text{ ha} \times 10,000 \text{ m}^2/\text{ha} / (10 \text{ m} \times 10 \text{ m})] \\ &= 105 \text{ seedlings/hectare}\end{aligned}$$

7. If the recommended rate of nitrogen is 90 kg/hectare, how many kilograms of urea fertilizer are to be applied per hectare using 46-0-0 urea fertilizer alone?

A. 195 B. 196 C. 198 D. 200

Answer: B

Solution:

Computation procedure: To determine the fertilizer rate for a particular nutrient, multiply the rate of the desired nutrient by 100 and divide by the percentage of the nutrient in the fertilizer (Manitoba Soil Fertility Guide, 2022).

$$\begin{aligned}\text{Fertilizer rate} &= (\text{Rate of Desired Nutrient} \times 100) / \text{Percentage of the Nutrient in the Fertilizer} \\ &= (90 \text{ kg/ha} \times 100) / 46 \\ &= 195.65 \text{ kg/ha}\end{aligned}$$

8. Based on the experimental field's soil analysis, the recommended fertilization rate for P_2O_5 is 45 kg/ha. What should be the fertilization application rate of P_2O_5 in kilograms/hectare if using 0-22-0 fertilizer alone?

A. 200 B. 202 C. 205 D. 207

Answer: C

Solution:

$$\begin{aligned}\text{Fertilizer rate} &= (\text{Rate of Desired Nutrient} \times 100) / \% \text{ of the Nutrient in the Fertilizer} \\ &= (45 \text{ kg/ha} \times 100) / 22 \\ &= 204.5 \text{ kg/ha}\end{aligned}$$

9. Based on the farm's soil analysis, the recommended fertilization rate for K_2O is 16.8 kg/ha. What should be the weight per unit area in kilograms/hectare of K_2O to be applied if using 0-0-60 fertilizer alone?

A. 28 B. 82 C. 2.8 D. 8.2

Answer: A

Solution:

$$\begin{aligned}\text{Fertilizer rate} &= (\text{Rate of Desired Nutrient} \times 100) / \% \text{ of the Nutrient in the Fertilizer} \\ &= (16.8 \text{ kg/ha} \times 100) / 60 \\ &= 28 \text{ kg/ha}\end{aligned}$$

Difficult Questions

10. If the recommended rate of nitrogen is 90 kg/hectare, how many bags of urea fertilizer are needed per hectare using 46-0-0 urea containing 50 kilograms per bag?

- A. 3 B. 4 C. 5 D. 6

Answer: B

Solution:

Computation procedure: To obtain the number of bags of fertilizer, divide the fertilizer requirement in kg/hectare by the number of kilograms contained per bag, then round up (not round off) to the nearest integer.

$$\begin{aligned}\text{Fertilizer rate} &= (\text{Rate of Desired Nutrient} \times 100) / \% \text{ of the Nutrient in the Fertilizer} \\ &= (90 \text{ kg/ha} \times 100) / 46 \\ &= 195.65 \text{ kg/ha}\end{aligned}$$

$$\begin{aligned}\text{No. of Bags} &= \text{Fertilizer Requirement in kg/hectare} / \text{No. of Kilograms Contained per Bag} \\ &= 195.65 \text{ kg/ha} / 50 \text{ kg/bag} \\ &= 3.91 \text{ bags/ha, round up to 4 bags/ha}\end{aligned}$$

11. Using a drone sprayer, you are to apply insecticide to a 10-hectare rice area. The label in the insecticide bottle states that application rate is: 1 liter of water per 20 ml of insecticide for every 100 square meters of rice field. How many liters of water and insecticide are needed to spray the entire area?

- A. 10,000 liters water, 20 liters insecticide
B. 20,000 liters water, 40 liters insecticide
C. 5,000 liters water, 10 liters insecticide
D. 30,000 liters water, 60 liters insecticide

Answer: A

Solution:

$$\begin{aligned}\text{Water vol.} &= \text{Area in m}^2 \times \text{Water mixing rate} \\ &= (10 \text{ has.} \times 10,000 \text{ m}^2/\text{ha}) \times (1 \text{ liter}/10 \text{ m}^2) \\ &= (100,000 \text{ m}^2) \times (1 \text{ liter}/10 \text{ m}^2) \\ &= 10,000 \text{ liters}\end{aligned}$$

$$\begin{aligned}\text{Insecticide vol.} &= \text{Area in m}^2 \times \text{Insecticide mixing rate} \\ &= (10 \text{ has.} \times 10,000 \text{ m}^2/\text{ha}) \times (20 \text{ ml}/100 \text{ m}^2 \times 1 \text{ liter}/1,000 \text{ ml}) \\ &= (100,000 \text{ m}^2) \times (0.0002 \text{ liter}/\text{m}^2) \\ &= 20 \text{ liters}\end{aligned}$$